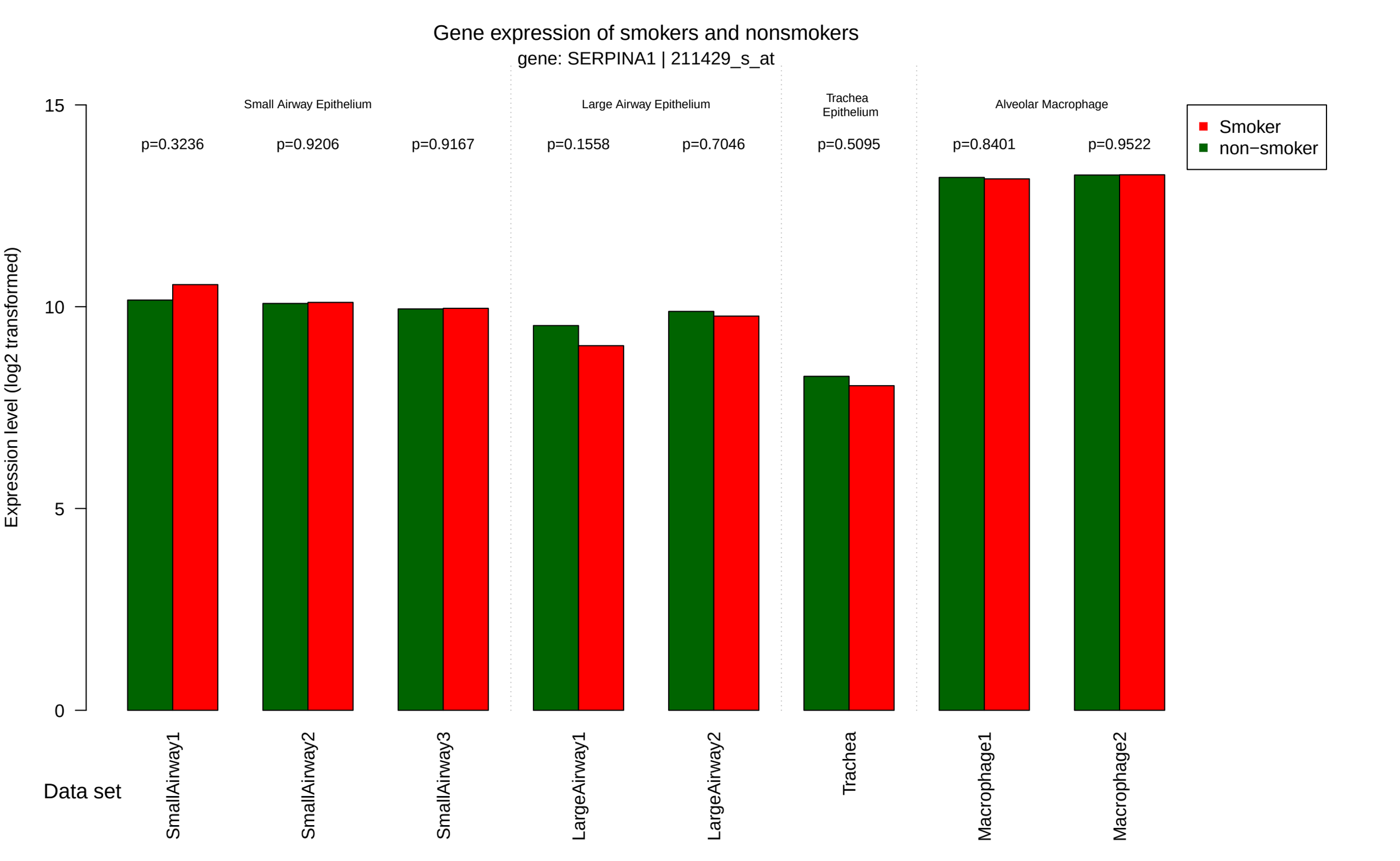
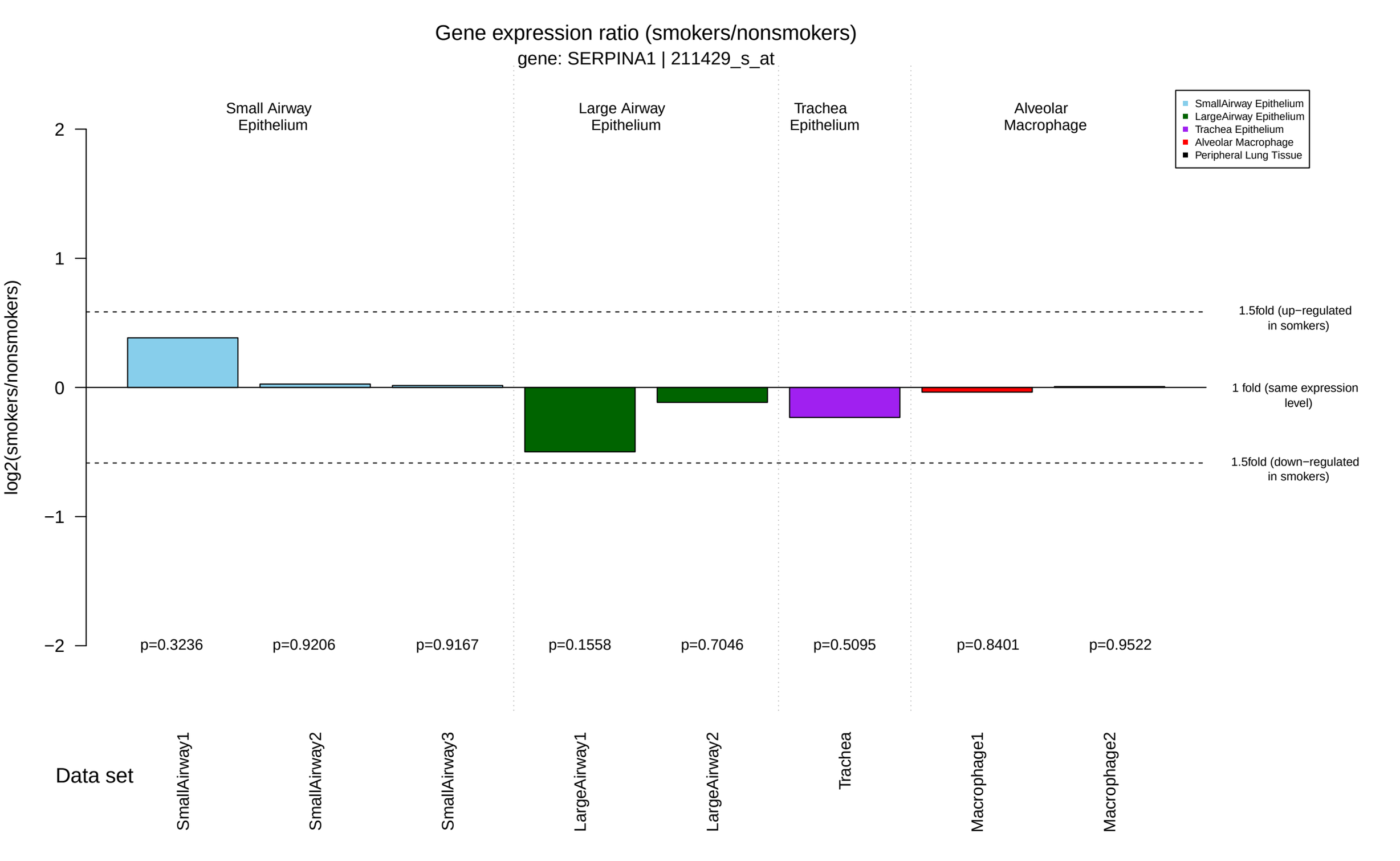
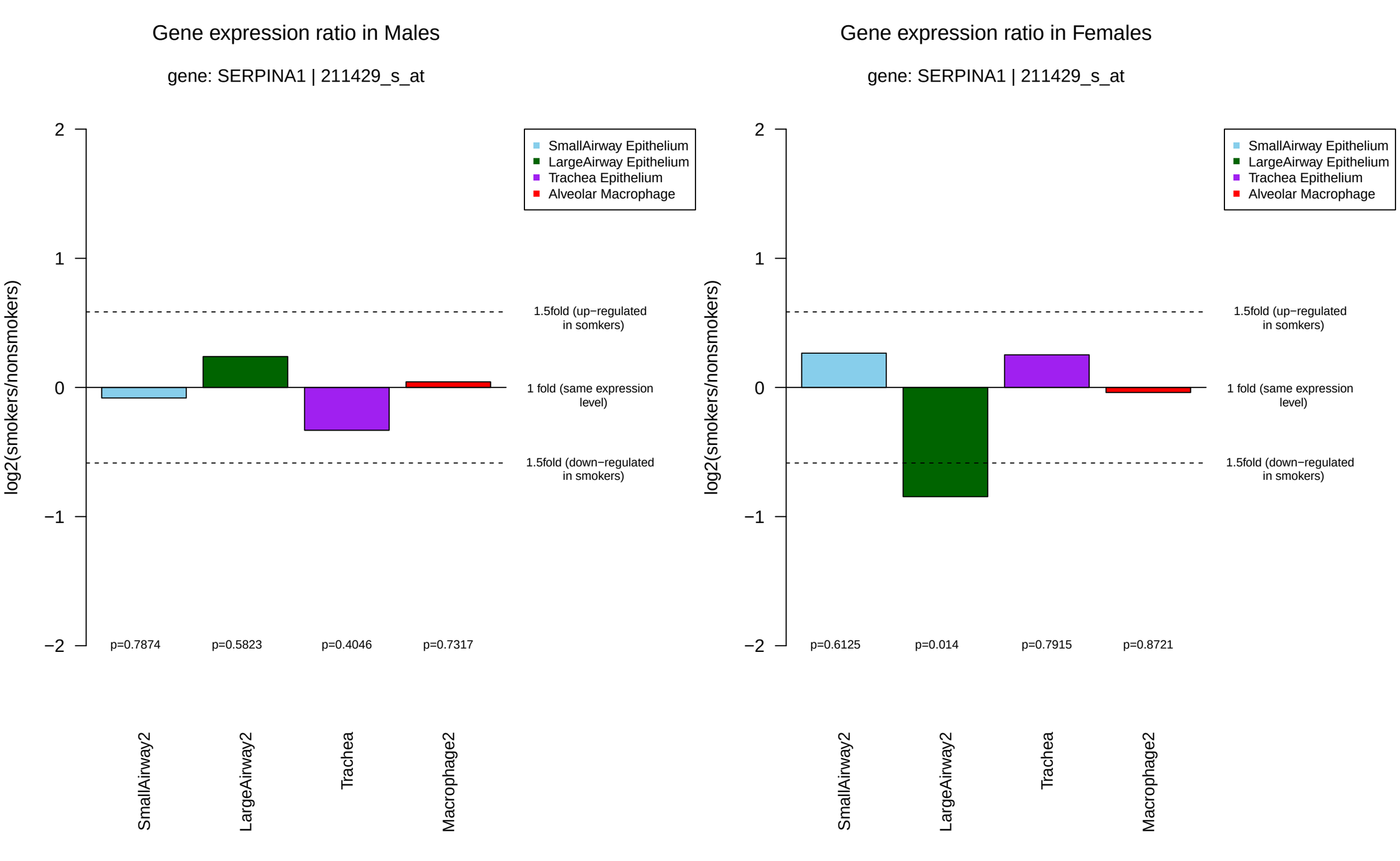
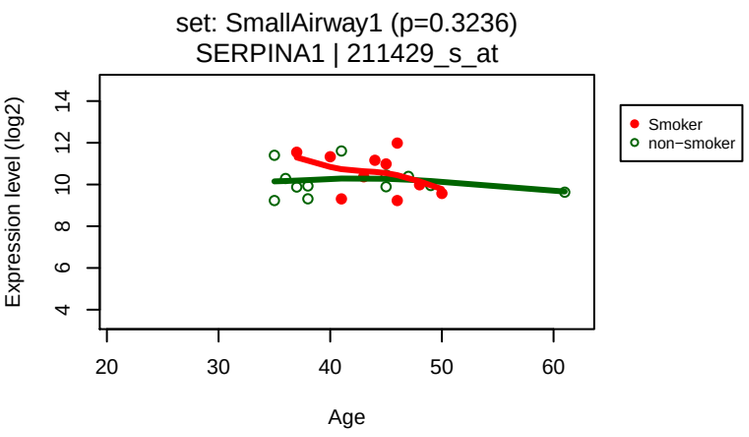
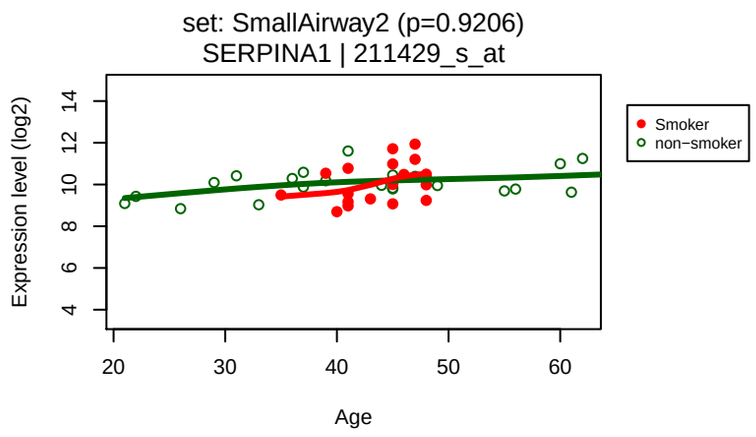
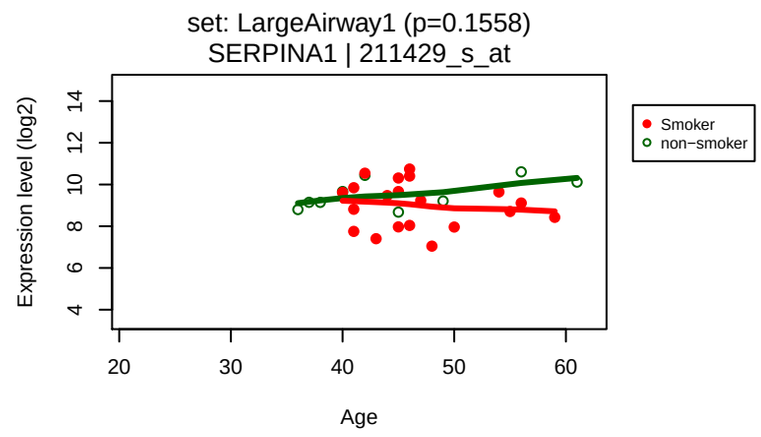

Gene expression of smokers and nonsmokers
gene: SERPINA1 | 211429_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3236
p=0.9206
p=0.9167
p=0.1558
p=0.7046
p=0.5095
p=0.8401
p=0.9522
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

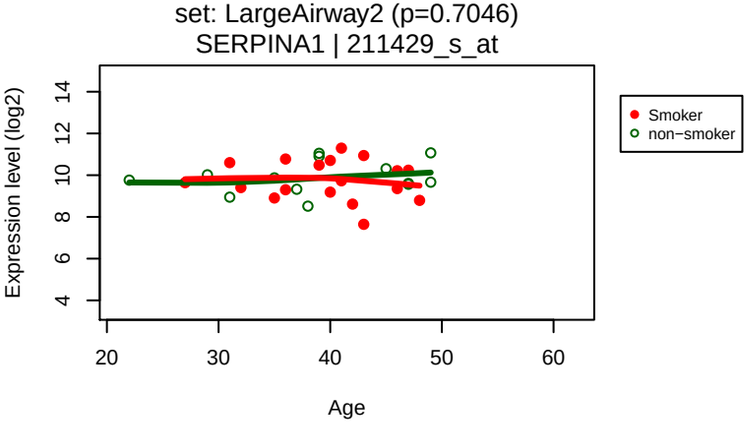

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA1 | 211429_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3236
p=0.9206
p=0.9167
p=0.1558
p=0.7046
p=0.5095
p=0.8401
p=0.9522
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

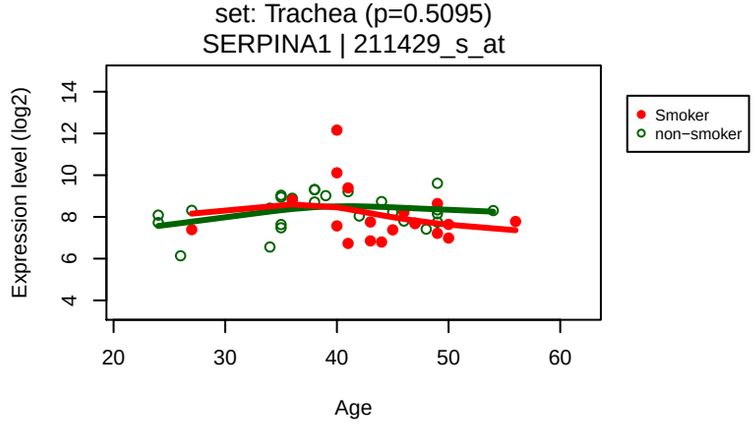

Gene expression ratio in Males
gene: SERPINA1 | 211429_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7874
p=0.5823
p=0.4046
p=0.7317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA1 | 211429_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6125
p=0.014
p=0.7915
p=0.8721
SmallAirway2
LargeAirway2
Trachea
Macrophage2

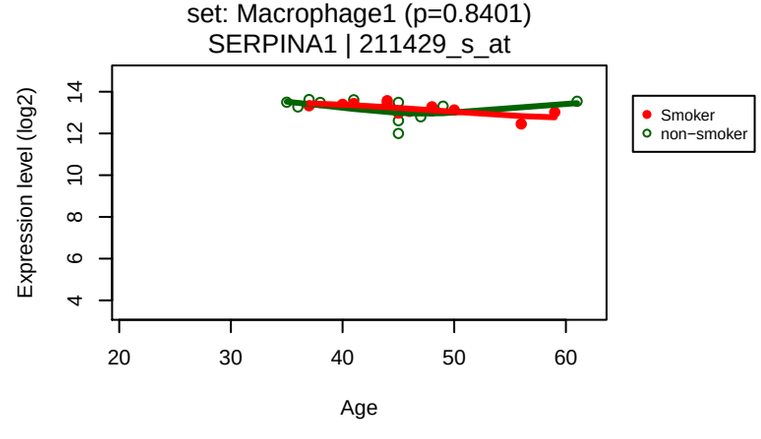

set: SmallAirway1 (p=0.3236)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker

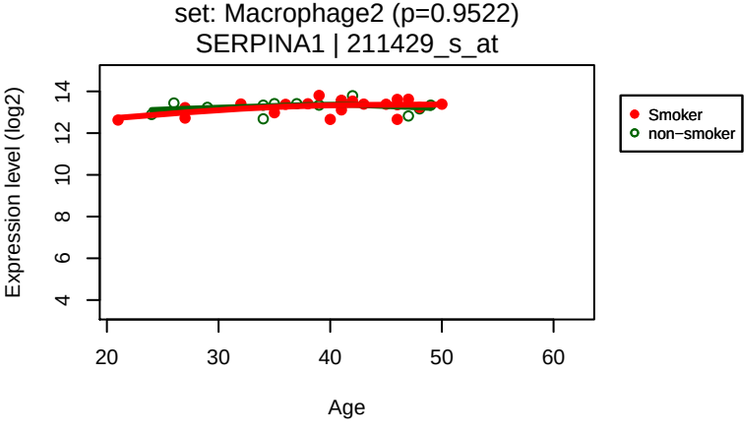

set: SmallAirway2 (p=0.9206)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker

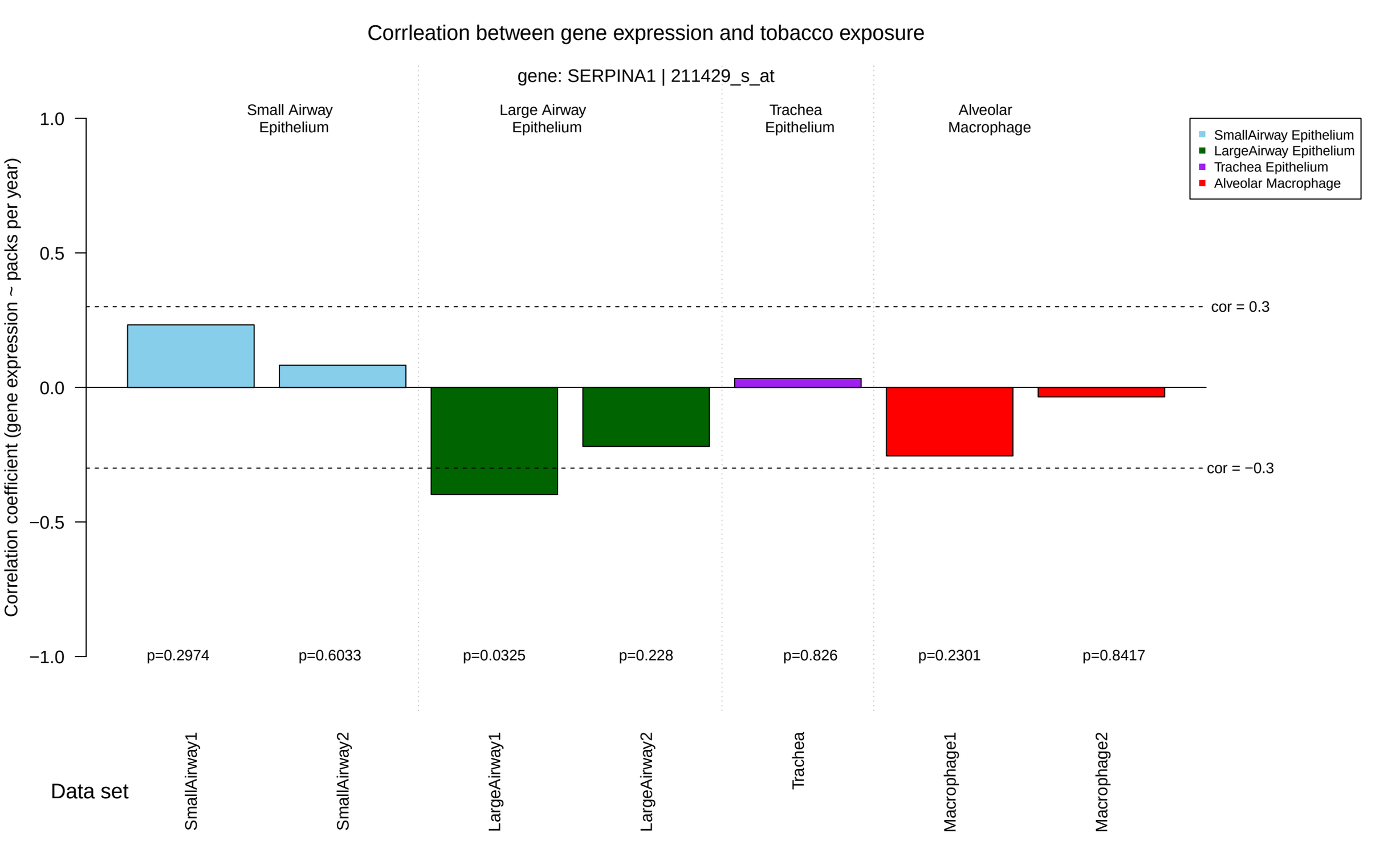

set: LargeAirway1 (p=0.1558)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7046)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5095)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8401)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9522)
SERPINA1 | 211429_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA1 | 211429_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2974
p=0.6033
p=0.0325
p=0.228
p=0.826
p=0.2301
p=0.8417
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2